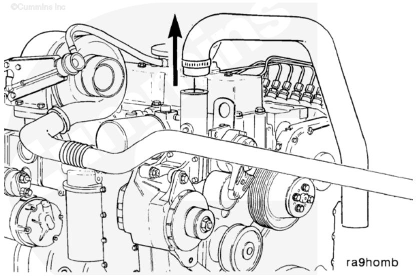
Trainings

Initial Check

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



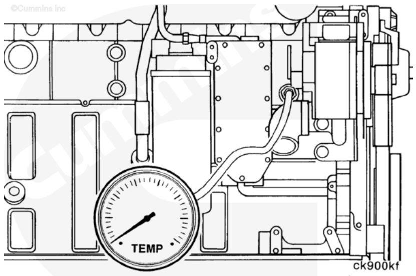
⚠ WARNING ⚠

Complete this test with the engine coolant temperature below 49°C [120°F]. Hot steam can cause serious personal injury.

Drain 2 liters [2.1 qt] of coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Remove the cooling system hose from the thermostat housing.

Install a thermocouple or temperature gauge, which is known to be accurate, in the ¾-inch pipe plug located at the front of the cylinder block.

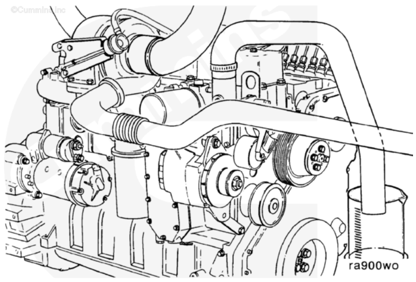


Install a hose of the same size on the thermostat housing outlet that is long enough to reach a remote, dry container used to collect coolant.

Install and tighten a hose clamp on the housing outlet.

Torque Value: 6 n•m [50 in-lb]

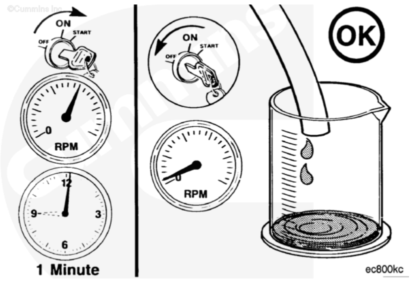
Insert the end of the hose in a dry container.



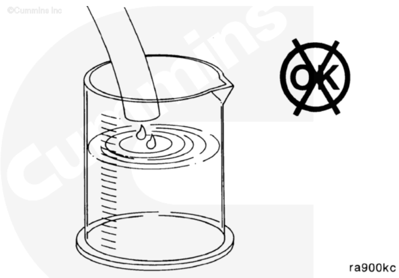
Operate the engine at rated speed for 1 minute.

Shut off the engine and measure the amount of coolant collected in the container.

ml		fl-oz
150	MAX	5



If more than 150 ml [5 fl-oz] of coolant is collected, the thermostats are leaking and **must** be replaced.



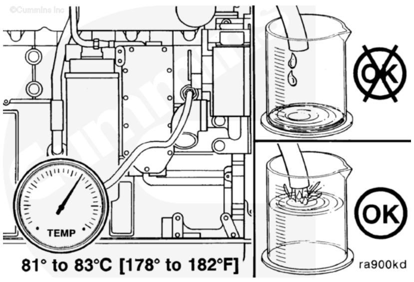
Complete the following in-chassis test to determine the thermostat opening temperature.

Start the engine and monitor the water temperature gauge and the container. The thermostat initial opening temperature is:

81 to 83°C [178 to 182°F]

Shut off the engine when the coolant starts to flow.

Note : If coolant does not start flowing into the container during the initial opening temperature range, the thermostat must be replaced.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Drain 2 liters [2.1 qt] of coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Remove the upper radiator hose from the outlet connection.

Remove the drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html).

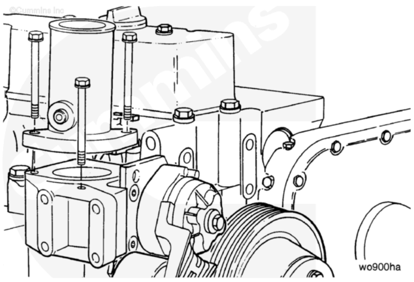
Disconnect the batteries.

Remove the alternator. Refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/100/100-013-001.html).

Remove

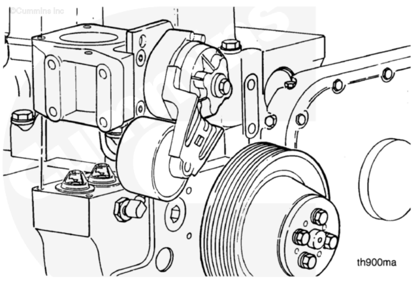
Remove the capscrews from the thermostat housing.

Remove the water outlet connection.



Remove the thermostat housing and belt tensioner assembly.

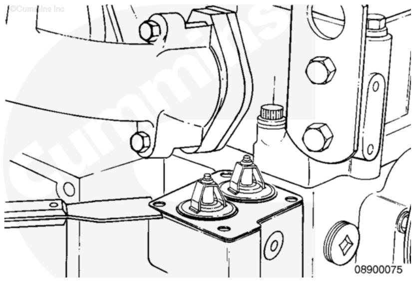
Note : If the vehicle is equipped with an external bypass system, the thermostat housing support (between the thermostat housing and cylinder block) must be removed.



⚠ CAUTION ⚠

Debris in the cooling system can cause damage to the engine.

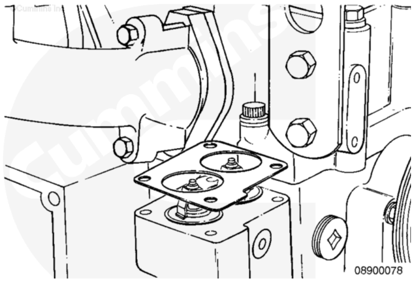
Remove the thermostat gasket and clean the gasket surface.



⚠ CAUTION ⚠

Do not shim the thermostats beyond the top of the block.

Measure the distance from the thermostat flange to the top of the block surface of each thermostat to determine the proper shim(s) to use.

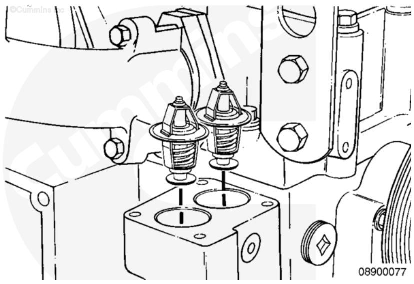


Note : The service shims included are 0.25 mm [0.010 in], 0.50 mm [0.020 in], 0.75 mm [0.030 in], and 1 mm [0.040 in].

Select the appropriate combination that will bring the thermostat height as close to the top of the block as possible.

Note : Any combination of shims can be used, but stacking is limited to a maximum of two shims per bore.

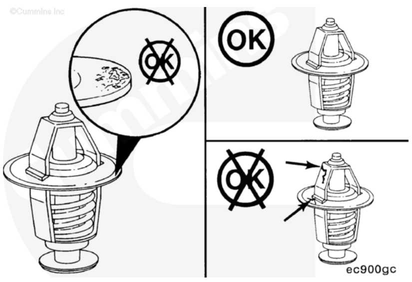
Remove each thermostat and insert the selected shims into each bore, making sure each shim is seated properly in the bore.



Inspect for Reuse

Inspect the thermostats for damage.

Make sure both thermostats are clean and free from corrosion.

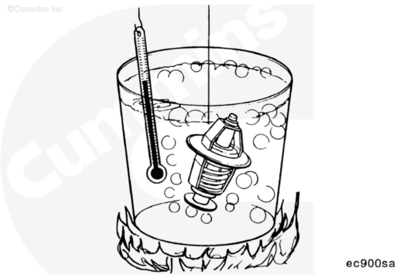


Suspend the thermostats and a 100°C [212°F] thermometer in a container of water.

Note : Do not allow the thermostats or thermometer to touch the container.

Heat the water slowly so the wax element in the thermostats has sufficient time to react to the rising water temperature.

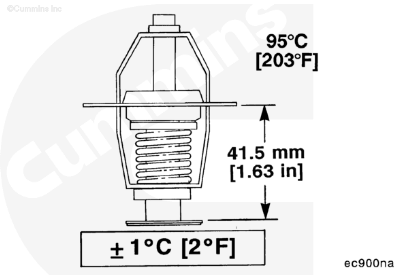
Note : The normal operating temperature is stamped on the thermostat.



Inspect the thermostats as follows:

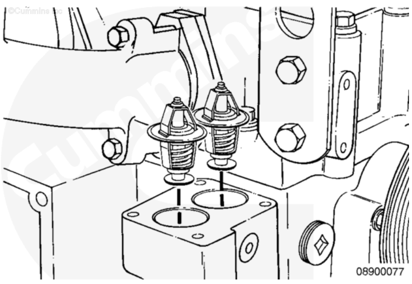
- Thermostat **must** begin to open within 1°C [2°F] of 82°C [180°F].
- Thermostat **must** be fully open within 1°C [2°F] of 95°C [203°F].

Note : The fully open clearance between the thermostat flow valve and flange must be 41.5 mm [1.63 in] minimum.

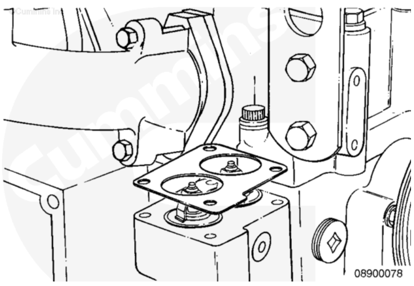


Install

Install the thermostats on top of the service shim(s) in the thermostat flanges. They can be within 0.23 mm [0.009 in] of flush with the top of the block, without being above the top of the block.

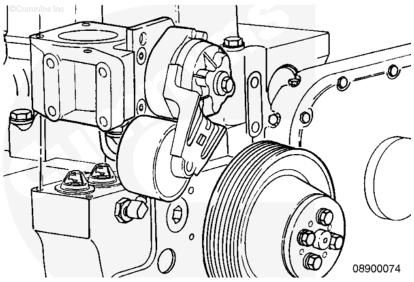


Install a new thermostat gasket.



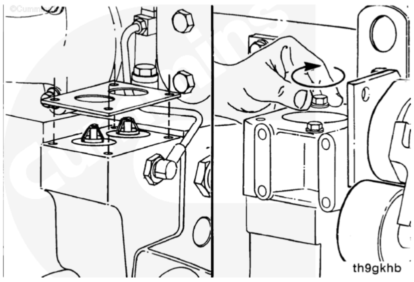
Position the thermostat housing and belt tensioner over the thermostats and gasket.

Note : If an external bypass system is used, the thermostat housing support (between the thermostat housing and cylinder block) must be installed.



Make sure the gasket is aligned with the capscrew holes.

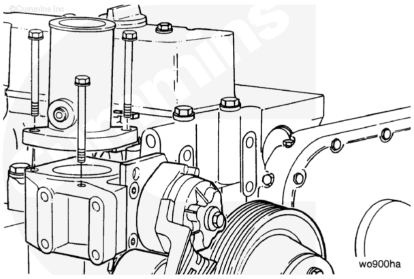
Install the capscrews and finger-tighten.



Install the water outlet connection.

Tighten all capscrews.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Install the alternator. Refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/100/100-013-001.html).

Install the drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html).

Install and tighten the battery's electrical connections.

Note : During filling, air must be vented from the engine's coolant passages. Open the engine vent

petcock, if equipped. Make sure to open the petcock on the aftercooler for aftercooled engines. The system must be filled slowly to prevent air locks. Wait 2 to 3 minutes to allow air to be vented; then add coolant to bring the level to the bottom of the radiator filler neck.

Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Operate the engine to normal operating temperature and check for leaks.

Last Modified: 02-Sep-2003